Time: 1 hrs. (65)

**B.Tech. V Semester (Computer Engg.) Examination -2018**

**CEN - 501 Computer Architecture Sessional – I**

Max. Marks: 15

**Note: Attempt All Questions**

| CO | | | |
|---|---|---|---|
| 1 | 1 | Describe Flynn's and Handler's classification scheme in detail with example. | 5 |
| 1 | 2 | Which of the addressing mode is most appropriate for the given process and why. Discuss?<br>(a) Array Implementation.<br>(b) Writing relocatable code.<br>(c) Passing array as parameter. | 3 |
| 2 | 3 | Explain Amdahl's law? Calculate the Speedup for it? If a program consists of 0.47 fractional parts of parallel code. What would be the speedup if the no of processors are 32? Suppose that the given application is run on a 48-processor machine. Then the expected performance improvement using Amdahl's law would be? | 4 |
| 2 | 4 | What is Cache Coherence Problem? How can we resolve cache coherence problem in multiprocessor system. | 3 |

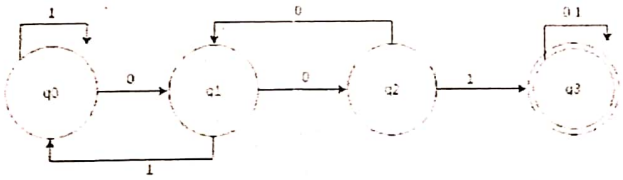
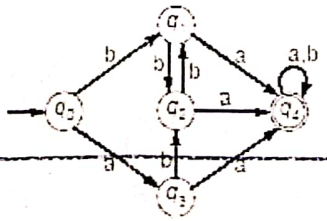
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B. Tech.( Computer Engineering) V Semester
1st Sessional 2018-19
AUTOMATA THEORY (CEN-502)
Time: 1 Hour MM:15

Note: Attempt all questions. Date : 17/09/18
Write your Roll no here .....

| Q No. | CO Mapped |
|---|---|
| 1 | 1 |
| 2 | 1 |
| 3 | 2 |
| 4 | 2 |
| 5 | 1 |

| Q. No | Question | Mark |
|---|---|---|
| 1. | Construct a DFA for the following expression: **0*((01)* + 2)*(1+0*2)** | 3 |
| 2. | With the help of a suitable explanation, find the minimum number of states required to recognize an octal number divisible by 3. | 3 |
| 3 | Using Arden's theorem, identify the Regular expression for the given DFA. The initial state is $q_0$.  | 3 |
| 4 | If L1 and L2 are two regular languages, then comment on the following statements:<br>**a)** L1 U L2 is not regular. **b)** L1 ∩ L2 is regular. | 3 |
| 5 | Minimize the following DFA using the Matrix Method:  | 3 |
| | **Validate your answers using suitable examples.** | |

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First Sessional Examination 2018

B.Tech. Computer Engineering V semester

**Microprocessor CEN-503**

65

Marks:-*15* Time:-One Hour

Attempt all questions. Marks are written in the margin.

Q1. Write a program in assembly language of 8085 to arrange a given set of integers stored at consecutive memory locations starting with address 2300H in descending order. First location contains number of item and following addresses contain items. Program should be well documented. 6 Marks

Q2. Write a program in assembly language of 8085 to covert 8-bit binary number into decimal number. Program should be well documented. 3 Marks

Q3. Write a program in assembly language using subroutine of 8085 for 8X8 multiplication by shifting multiplicand. Program should be well documented. 3 Marks

Q4. Write a program in assembly language of 8085 for finding 16 bit sum of only even parity integers out of 300 8-bit integers stored at consecutive locations. Program should be well documented. 3 Marks

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| Object Oriented Programming (CEN-504) | B.Tech (Computer Engineering) Semester –V Sessional – I 2018<br>Department of Computer Engineering<br>Jamia Millia Islamia | Time : 1 Hour<br>Max Marks: 15 |
|---|---|---|

65

| | | | |
|---|---|---|---|
| Q1. | CO1<br>CO2<br>CO3 | What is the difference between<br>(i) Polymorphism and Data Hiding<br>(ii) A copy constructor and an overloaded = operator<br>(iii) Function overriding and Function overloading | 3 |
| Q2. | CO1 | Define a class Person which has two data members: name (char*) and age (int). Define the following constructors (a.) default (b.) with name as argument (assume age is 18) (c.) with name and age as argument. Use new and delete for memory management. Also write a copy constructor to copy an object to another. Show when the copy is done by invoking a copy constructor and when the copy is done without invoking a copy constructor. | 3 |
| Q3. | CO2 | List the operators which cannot be overloaded at all and the operators which cannot be overloaded using friend function? Show how can you overload "<<" and ">>" operators to read and display the objects of a Matrix class. | 3 |
| Q4. | CO3 | With the help of a complete program demonstrate the concept of virtual function. | 3 |
| Q5. | CO3 | What do you mean by a parameterized class? Write a program to sort an array of anonymous data type T. Use a generic function swap () to swap the array elements. | 3 |

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B.Tech. (3rd year) V Semester (Computer Engg.) Examinations

Time: 1 hrs. **Computer Networks (CEN - 505) Sessional-I** Max. Marks: 15

**Note: Attempt All Questions**

| | | |
|---|---|---|
| 1 | The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are<br>(a) TCP, UDP, UDP and TCP<br>(b) UDP, TCP, TCP and UDP<br>(c) UDP, TCP, UDP and TCP<br>(d) TCP, UDP, TCP and UDP<br>Justify your answer with explanation. | 3 |
| 2(i) | What is the need of error control at the data link layer when the transport layer provides error control? What is the difference between the two error controls? | 2 |
| (ii) | If the unit exchanged at the data link layer is called a frame and the unit exchanged at the network layer is called a packet, do frames encapsulate packets or do packets encapsulate frames? Explain your answer. | 2 |
| 3 | An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to three groups of customers as follows:<br>a. The first group has 32 customers; each needs 256 addresses.<br>b. The second group has 128 customers; each needs 128 addresses<br>c. The third group has 128 customers; each needs 32 addresses.<br>Design the sub blocks and find out how many addresses are still available after these allocations. | 4 |
| 4 (i) | Assume a data stream as 111001110101. Encode this stream using the following:<br>a. NRZ-I<br>b. Manchester Encoding<br>c. Differential Manchester Encoding | 3 |
| (ii) | Assume a data stream as 1000000000110. Encode this stream using HDB3 Encoding | 1 |

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6).

B. Tech.( Computer Engineering) V$^{th}$ Semester
2$^{nd}$ Sessional 2018-19
AUTOMATA THEORY (CEN-502)
Time: 1 Hour MM:15

**Note: Attempt all questions.** **Date : 12/11/18**

**Write your Roll no here :**

| Q No. | CO Mapped |
|---|---|
| 1 | 2 |
| 2 | 3 |
| 3 | 3 |
| 4 | 4 |

| Q. No | Question | Marks |
|---|---|---|
| 1. | State and Explain the Pumping Lemma for regular languages. Using this Lemma. comment about the regularity of the following languages with suitable explanations :<br>i) $L1 = \{ ww \mid w \in \{0, 1\}^* \}$<br>ii) $L2 = \{ w \mid \#0(w) != \#1(w), w \in \{0, 1\}^* \}$ | 4 |
| 2. | Construct the CFG for the following language :<br>$L3 = \{a^n b^n c^m d^m \mid n, m>0 \} \cup \{a^n b^m c^m d^n \mid n, m > 0 \}$<br>Prove whether the CFG for this language is ambiguous or not. | 3 |
| 3 | Convert the following CFG to CNF along with mentioning appropriate rules wherever needed<br>$S \rightarrow ASB \mid AB$<br>$A \rightarrow aAS \mid a \mid \varepsilon$<br>$B \rightarrow SbS \mid AB \mid bb$ | 5 |
| 4 | Define the Push Down Automaton. Construct a Deterministic PDA for the Language accepting set of all strings over {0, 1} which do not contain the substring 010. | 3 |

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| Object Oriented Programming (CEN-504) | **B.Tech (Computer Engineering) Semester –V Sessional – II 2018**<br>**Department of Computer Engineering**<br>**Jamia Millia Islamia** | Time : 1 Hour<br>Max Marks: 15 |
|---|---|---|

| | | | |
|---|---|---|---|
| Q1. | CO4 | A super class reference can refer to a subclass object. Write a program in Java to demonstrate this concept. Your code must be well commented. | 2 |
| Q2. | CO4 | Describe the use of following keywords in JAVA with suitable example programs.<br>(i) final<br>(ii) super<br>(iii) static | 6 |
| Q3. | CO5 | List the keywords associated with exceptions in Java. Give one sentence description of what each one is for and a piece of code to demonstrate its usage. | 5 |
| Q4. | CO5 | Given:<br><br>`class Uber {`<br>`  static int y = 2;`<br>`  Uber(int x) { this(); y = y * 2; }`<br>`  Uber() { y++; }`<br>`}`<br>`class Minor extends Uber {`<br>`  Minor() { super(y); y = y + 3; }`<br>`  public static void main(String [] args) {`<br>`  new Minor();`<br>`  System.out.println(y);`<br>`} }`<br>What is the result?<br><br>A. 6<br>B. 7<br>C. 8<br>D. 9<br>E. Compilation fails.<br>F. An exception is thrown. | 2 |

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Second Minor Examination 2018
B.Tech. (Computer Engineering) V semester
Microprocessor

Time:-One Hour

Max Marks:-15

Attempt all questions. All questions carry equal marks

Q1. How is 8086 different from 8085 ? Discuss in details.

Q2. Draw and explain timing diagram of LDA 2500h

Q3.Explain how to implement interrupt for multiple peripheral using INTR with the help of suitable example and diagram.

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Time: 1 hr

65

B.Tech. 5th Semester (Computer Engg.) Examination -2018

CEN - 501 Computer Architecture Sessional – II

**Note: Attempt All Questions**

Max. Marks: 15

| Ques | CO-Mapped |
|---|---|
| Q1 | 3 |
| Q 2 | 3 |
| Q 3 | 5 |

Q1 Design 8-bit CLA and compare it with 8-bit CPA. Also calculate the worst case add time for 64 bit CLA.

Q 2 Draw the flow chart for Non-restoring algorithm. Divide the 22 by 5 using it and assume appropriate bits.

Q 3 Consider a 4 stage pipeline processor. The number of cycles needed by the four instructions I1, I2, I3, I4 in stages S1, S2, S3, S4 is shown below:

| | S1 | S2 | S3 | S4 |
|---|---|---|---|---|
| I1 | 3 | 1 | 2 | 1 |
| I2 | 2 | 3 | 2 | 2 |
| I3 | 1 | 2 | 1 | 3 |
| I4 | 2 | 2 | 3 | 2 |

What is the number of cycles needed to execute the following loop?
For (i=1 to 2) {I1; I2; I3; I4;}? Solve it using phase time diagram? Calculate the speedup?

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CEN-505

Your Roll No. ...16BCS065

**B. Tech. (Computer Engineering)**

**V Semester Examination- 2018**

**COMPUTER NETWROKS**

**Paper No.: CEN-505**

**Time: 3 Hours** **MM: 60**

(*Write your Roll No. on the top immediately on receipt of this question paper.*)

**Note:** Attempt all questions. Make suitable assumption(s) wherever required.

**Q. 1** (a)(i) Differentiate the following with respect to OSI layers functionality: (3+3)
**CO1**

- Logical Address
- Physical Address
- Port Address

(ii) With the aid of an example, explain why subnetting was introduced? Determine the sub network address if the destination address is 143.112.115.83 and the subnet mask is 255.255.240.0?

(b)(i) If the unit exchanged at the data link layer is called a frame and the unit exchanged at the network layer is called a packet, do frames encapsulate packets or do packets encapsulate frames? Explain your answer. (3+3)

(ii) List two advantages and two disadvantages of having international standards for network protocols.

**OR**

(b') How is bridge different from repeaters? What are the advantages and disadvantages of each? State with reason, whether a LAN can be extended to any size by increasing the number of repeaters or not? (6)

**Q. 2** (a) Differentiate between bit rate and baud rate. What is the baud rate of Manchester encoding? Sketch the Manchester encoding and Differential Manchester encoding for the following bit stream: 1001111001. (6)
**CO2**

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(b) Explain Guided transmission media. Mention the characteristics that distinguish optical from twisted pair and coaxial cable. (6)

**OR**

(b') What is the essential difference between message switching and packet switching? Using a combination of time division and space division switching, design a circuit with 12 inputs and 12 outputs. The input multiplexers should be 4x1 and the output multiplexers should be 1x4. (6)

**Q. 3** (a) Discuss the services provided by the data link layer to the network layer? In a data link protocol, the frame delimiter flag is given by 0111. Assuming the bit stuffing is employed and the data sequence is 01110110, determine the transmitted sequence after bit stuffing? (6)

**CO3**

(b) A 3000 km long trunk is used to transmit frame using a Go back N protocol. The propagation speed is 6 μsec / km and trunk data rate is 1.544 Mbps. The frame size is 64 bytes and we ignore the transmission time taken by the receiver for acknowledgement. If Go back N protocol is used in order to achieve efficiency of 100%, then what is the maximum size of window at sender side. (6)

**OR**

(b') A 12 bit even parity hamming code whose hexadecimal value is E4F arrives at a receiver. What was the original value in hexadecimal? Assume that not more than 1 bit is in error. (6)

**Q. 4** (a) What is the hidden station problem and the exposed station problem? In wireless LAN, the point coordination function and distributed coordination function can coexist within one cell. Explain how it is possible to have central control and distributed control operating at the same time. (6)

**CO4**

(b) What is meant by vulnerable period? Show that the vulnerable time period of slotted ALOHA is half of the pure ALOHA? A group of N stations share a 50-kbps slotted aloha channel. Each station outputs 500 bits from on an average of once every 5000 msec, even if previous one has not been sent. What is the maximum value of N? (6)

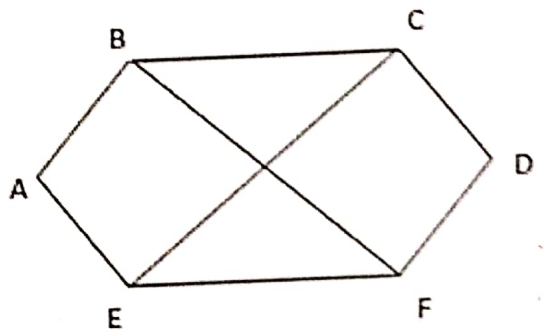
**OR**

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(b') Define CSMA/CD and CSMA/CA. A network using CSMA/CD has a bandwidth of 10Mbps. If the maximum propagation time (including the delays in the devices) is 25.6 μsec. What is the minimum size of the frame? (6)

Q. 5 CO5

(a) Discuss the problem of count to infinity associated with distance vector routing technique. For the following figure, the network uses a Distance Vector Routing protocol. The following vectors have just come into router C: from B: (5,0,8,12,6,2); from D: (16,12,6,0,9,10); and from E: (7,6,3,9,0,4). The measured delays to B, D and E are 6, 3 and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the expected delay (6)




(b) What are the reasons for congestion in a network? A computer on a 6-Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 1 Mbps. It is initially filled to capacity with 8 megabits. How long can the computer transmit at the full 6 Mbps. (6)

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BCEN-503 Your Roll No. .....16BCS065.....

# B.Tech.(Computer Engineering) V Semester Examination 2018
## Microprocessor
## Paper No. : CEN-503

**Time: 3 Hours** **Maximum Marks: 60**

(Write your Roll No. on top immediately on receipt of this question Paper)
Note: - Attempt all questions.

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**CO1**

Q1. (a). Draw and explain timing diagram of STA 2020 H instruction of 8085 stored at address 2300H. (06)

(b). Discuss in details how microprocessor 8086 is different from Microprocessor 8085. (06)

**CO2**

2(a). Discuss the addressing modes of 8086 with the help of suitable examples. (06)

2(b) Explain the meanings of the following instructions of 8085/8086 after writing their syntax. Also illustrate their use. (06)

(i) STAX

*(ii) SHLD*

*(iii) LOOP*

(iv) MOVSB

**CO3**

3 (a). Write a well documented program in assembly language of 8085 for 8X8 multiplication by without shifting partial product. (06)

3 (b). Write a well documented program for finding factorial of 8bit number. (06)

**OR**

3'(a) Write a well documented program in assembly language of 8085 to implement bubble sort algorithm. A given set of items stored at consecutive memory locations should be sorted in ascending order. 06

3'(b) Write a well documented program in assembly language of 8085 to convert 8 bit Binary number into decimal Number. 06

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**CO4**

4 (a). Explain how I/O devices are interfaced as I/O mapped I/O with help of suitable diagram and examples.. 06

4 (b). Discuss the interrupts available in 8085. 06

**CO5**

5. (a). Write a program in assembly language of 8086 to find out the product of two 3x3 matrices. 06

5(b). Describe how assembly language is used with C programs with help of suitable example. also discuss how the software interrupts programming are done in 8086. 06

**OR**

5' (a) Interface an 8255 with 8086 to work as an I/O port. Initialize port A as output port, port B as input port and port C as output port. Port A address should be 0740 H. Write a program to sense switch positions SW0- SW7 connected at port B , to display the sensed pattern on port A to which 8 LEDs are connected and to display the number of ON switches out of eight switches on port C lower ? (12)

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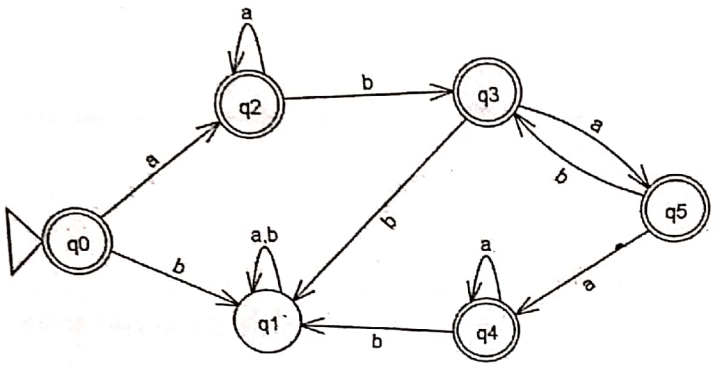
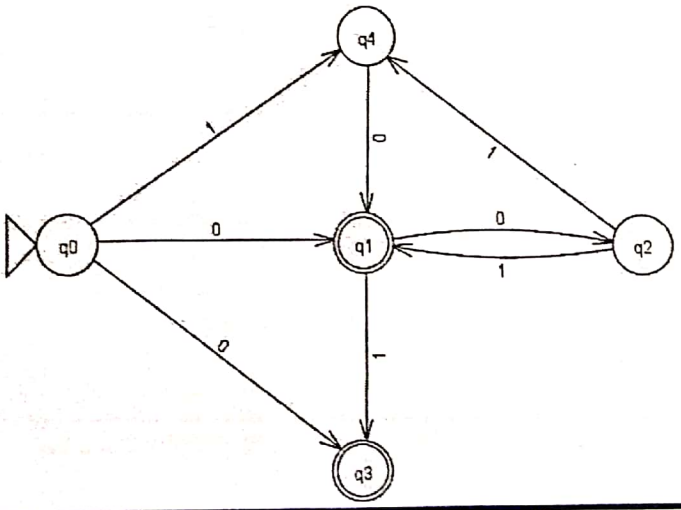
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# B.Tech. Comp. Engg. Vth Sem Examination 2018
## AUTOMATA THEORY
## Paper no CEN-502

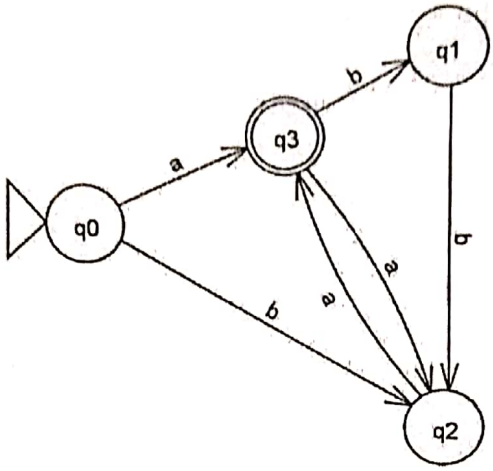
**Time : Three hours** **Max. Marks: 60**

*Attempt any TWO parts from each question.*

*Write your Roll No here* 16BCS5065 *on receipt of the question paper immediately*

| Q. No | CO Mapped | Questions | Marks |
|---|---|---|---|
| Q1 a) | CO1 | Construct a DFA for the followings and validate your answers:<br>i) 1* (10 + 010)*01<br>ii) ab(b*a + a*b)ba | 6 |
| Q1 b) | CO1 | Minimize the following DFA with matrix method only and verify your answer:<br>  | 6 |
| Q1 c) | CO1 | *Let w be any string of length n is {0,1}*. Let L be the set of all substrings of w. Explain in details about, what is the minimum number of states in a non-deterministic finite automaton that accepts L? Also Convert the following NFA to DFA and also verify your answer.*<br>  | 3 + 3 |
| Q2 a) | CO2 | Construct a Regular Expression for the following languages over {0,1} :<br>i) The set of all strings containing the substring 00.<br>ii) The set of all strings containing at most two 0's.<br>iii) The set of all strings containing at least two 0's. | 6 |

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| | | | |
|---|---|---|---|
| Q2 b) | CO2 | Using Pumping Lemma for Regular Languages, comment about the regularity of the following languages with suitable explanations :<br>i) $L1 = \{ a^n b^\ell a^k \mid n>5, \ell>3, k<=\ell \}$<br>ii) $L2 = \{ bba(ba)^n a^{n-1} \mid n>=0 \}$ | 6 |
| Q2 c) | CO2 | State and prove the Arden's theorem. Also construct the RE for the given DFA:<br>  | 6 |
| Q3 a) | CO2 | Explain the Languages of the following CFGs<br>i) G1 : S→aS \| A  A → b \| bA<br>ii) G2 : S → aA \| bB  A → aA \| B \| ε  B→ bB\| ε | 6 |
| Q3 b) | CO3 | Construct a CFG for a language accepting all strings over {a, b} which do not contain '*aab*'. Also identify the language of the given grammar:<br>S → AbB<br>A → aA \| a<br>B → bB \| b | 6 |
| Q3 c) | CO3 | Convert following CFG to CNF<br>P → RB \| aRBa  B → Bb \| aTbT<br>R → aT \| TT  T → d \| ε<br>Mention all the steps and the corresponding rules used at appropriate positions | 6 |
| Q4 a) | CO4 | Mathematically define a Push Down Automaton. Also present the description of all its components in details. | 6 |
| Q4 b) | CO4 | Construct a PDA for the followings:<br>i) $L1 = \{ w \in \{0,1,2\}^* \mid w$ does not equal $xcx$ for some $x \in \{0,1\}^* \}$.<br>ii) $L1 = \{ w \in \{0,1\}^* \mid \#_0(w) != \#_1(w) \}$.<br>Validate your answers | 6 |
| Q4 c) | CO4 | Construct a PDA with final state acceptance for the language **B** defined as:<br>**B = { bin(i)$mir(bin(i + 1)) \| i > 0 } ⊆ {0; 1; $}* }**<br>Here *mir(i)* ∈ {0; 1}* is the mirror function and *bin(i)* ∈ {0; 1}* the binary representation (without leading zero's) of the number *i*. Eg. bin(11) = 1011 and mir(bin(12)) = 0011 | 6 |

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| | | | |
|---|---|---|---|
| Q5 a) | CO5 | Design a TM for the operation: f( n, m) = n ÷ m. Verify your answer with suitable examples. | 6 |
| Q5 b) | CO5 | What are primitive recursive functions? Show that $f(x, y) = 2x^2 - y$ is primitive recursive. | 6 |
| Q5 c) | CO5 | Explain following terms in details with suitable examples:<br>i) Multi tape Truing Machine<br>ii) Recursive Vs Recursively Enumerable Languages<br>iii) NP Completeness | 6 |

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CEN-504

Roll No. 65

**B.Tech. (Computer Engineering) V Semester Examination, 2018**

**Object Oriented Programming**
**Paper No. CEN-504**
**Time: Three Hours**
**Maximum Marks: 60**

(Write your Roll No. on the top immediately on receipt of this question paper)
**Note: Attempt all questions. Make suitable assumptions wherever necessary.**

**Q.1 CO1** **(a)** Define the term *Object Oriented Programming* briefly. Write a program in C++ to declare a class called *book_details* to represent details for a book, having data members like *title*, *author*, *edition*, *price* and *no_of_copies_available*. Define following functions: - *constructor(s)* – *display* function to display all data members - *find_books* to find and display details of all books having price less than Rs. 250 – main to create an array of book_details and to show usage of above functions. **6**

**(b)** Describe the following terms In C++ with example. **6**
**(i)** *Default Arguments* **(ii)** *Reference variable* **(iii)** *Destructor*
**(iv)** *this pointer* **(v)** *Inline functions* **(vi)** *Static Block*

**Q.2 CO2** **(a)** Declare a class called *bird* having private data members *name* and *weight*. Define following functions: - *default constructor* for reading data members from key board - *overloaded constructor* with two arguments to be used for initialization of data members. - *display* function to display data members. - overloaded member *operator* >= to compare weight of two bird objects, returning false if weight of first bird object is less than that of the second & true otherwise. Define main to illustrate use of above functions. **6**

**(b)** Define the term *Generic Programming*. Create a generic class *stack* using template and implement common Push and Pop operations for different data types. **6**

**OR**

**Q.2' CO2** **(a)`** Explain the term *operator overloading*. Create a class *coordinate* containing x, y and z private variables. Perform operations for incrementing, adding and comparing object(s) by overloading + +, + = and = = operators respectively. Define necessary functions to set and display the variables. **6**

**(b)`** What is the difference between *class template* and *function template*? Declare a template class called *exam* having an array of generic type as a data member, named *elements*[10]. Define following generic (template) member functions: - sort( ) to arrange elements in ascending order - *find_max*( ) to find and return maximum from the array. Define *main* to illustrate usage of these functions to process two different types of data. **6**

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| | | | |
|---|---|---|---|
| **Q.3 CO3** | **(a)** | Declare a class called *item* having data members *item_code*, *item_name*, *cost* and *discount*. Derive two classes from class item, namely *employee* and *customer*. The class employee has data members like *employee_code*, *employee_name* and *amount*. The class customer has data members like *customer_name* and *amount*. Define functions for - initializing data members. - displaying the values of data members. - computing amount to be paid for a purchased item. Also define function main to create objects of both derived classes and to show usage of above functions | **6** |
| | **(b)** | What is the difference between compile time and run time polymorphism? Explain the usage of virtual keyword in implementing run time polymorphism with the help of a suitable example. | **6** |
| | | **OR** | |
| **Q.3' CO3** | **(a)`** | What do you mean by Inheritance? What are its different types? Explain the concept of virtual base class with a suitable program in C++. | **6** |
| | **(b)`** | Demonstrate how parameters are passed to base class constructors in multilevel inheritance in C++. Also discuss the order of execution of constructors and destructors. | **6** |
| ***Q.4 CO4*** | **(a)** | What are main features of an abstract class in JAVA? Implement an abstract class called Shape which has three subclasses say Triangle, Rectangle, and Circle. Define one method area() in the abstract class and override this area() in these three subclasses to calculate for specific object i.e. area() of Triangle subclass should calculate area of triangle etc. Same for Rectangle and Circle. Which OOP feature is implemented by this program? | **6** |
| | **(b)** | Explain interface in JAVA. Write a program that illustrates interface inheritance. Interface P is extended by P1 and P2. Interface P12 inherits from both P1 and P2.Each interface declares one constant and one method. class Q implements P12.Instantiate Q and invoke each of its methods. Each method displays one of the constants. | **6** |
| **Q.5 CO5** | **(a)** | Discuss the mechanism of keywords try, catch and finally to handle Exceptions in Java. Explain with an example like "division by zero" situation? Show with example working of *Nested Try* and *multiple catch* blocks. | **6** |
| | **(b)** | What do you mean by a Thread? Write a program to create multiple Threads in JAVA using Runnable interface. Why synchronization is required in multithreaded programming and how can we implement it in program? | **6** |

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CEN-501 Roll No.....16BCS5065.....

**B. Tech. Computer Engineering 5th Semester Examination, 2018**
**COMPUTER ARCHITECTURE**
**Paper No. CEN-501**

**Time: Three Hours** **MM: 60**

(Write your Roll No. on the top immediately on receipt of this question paper)

**Note: Attempt All Questions. Assume suitable missing data, if any.**

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|---|---|---|---|
| **Q1** [CO1] | a) | Explain Shore's and Flynn's Computer architectural classification scheme in details with example. | 6 |
| | | **OR** | |
| | a)' | What are the elements of instruction set Architecture? | 6 |
| | b) | State Amdahl's Law. Using Amdahl's Law, derive the speedup of a processor after the application of enhancement. If a program consists of 0.32 fractional parts of parallel code. What would be the speedup if the no of processors are 32? Suppose that the given application is run on a 64-processor machine. Then the expected performance improvement using Amdahl's law would be? | 6 |
| **Q2** [CO2] | a) | Design 16-bit adder using 4-bit CLA and compare it with 16-bit adder using CPA. Also calculate the worst case add time for both. What do you understand by Carry Save Adder? | 6 |
| | b) | Draw the flow chart for Non-Restoring Division algorithm. Divide 36 by 8 using it and assume appropriate number of bits for numbers. | 6 |
| | | OR | |
| | b)' | (i) Represent 28.75 in IEEE 754 standard for representing floating numbers in single and double precision format. What is range of single and double precision format? | 4+2=6 |
| | | (ii) Which of the addressing mode is most appropriate for the following given process and why. Discuss?<br>▪ Array Implementation.<br>▪ Program Relocation | |
| **Q3** [CO3] | a) | Design a 4 X 4 Array Multiplier using AND gates and Full Adder. What would be the time taken to multiply two n-bit number? | 6 |
| | | OR | |
| | a)' | Draw the flow chart for Booth multiplication algorithm and show all the steps for multiplication of (-38) and (9) using it. (Take appropriate number of bits). | 6 |

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b) Design a hardwired controller for calculating the sum of following series up to N terms: 6

$1 - 2 + 3 - 4 + 5 - 6 + \ldots\ldots\ldots\ldots N.$

Q4 [CO4] a) Explain different types of shared memory systems with example. Explain the NUMA model in details. Is NUMA tightly coupled system? 6

OR

a)' (i) Explain Structural and Control Dependency with example? 4+2=6

(ii) Consider a 5 stage pipeline processor. The number of cycles needed by the four instructions I1, I2, I3, I4 in stages S1, S2, S3, S4 is shown below:

| | S1 | S2 | S3 | S4 | S5 |
|---|---|---|---|---|---|
| I1 | 2 | 1 | 1 | 1 | 2 |
| I2 | 1 | 3 | 2 | 2 | 1 |
| I3 | 1 | 1 | 1 | 3 | 2 |
| I4 | 2 | 2 | 3 | 2 | 1 |

What is the number of cycles needed to execute the following loop? For (i=1 to 2) {I1; I2; I3; I4; I5 ;}? Solve it using phase time diagram? Calculate the speedup?

b) What is Cache coherence Problem? How can we resolve the cache coherence problem in Multiprocessor systems? Explain different types of protocols for it. 6

Q5 [CO5] a) What is the difference between a scalar and a vector processor? What are the advantages of a vector processor over other parallel processor? 6

b) (i) Explain the dynamic branch prediction state diagram?

(ii) A 5-stage pipeline has the stage delays as 180, 150, 120, 160 and 140 nanoseconds respectively. Registers that are used between the stages have a delay of 4 nanoseconds each. Assuming constant clocking rate, what will be the total time taken to process 100 data items on this pipeline? 3+3=6

OR

b)' Explain the following briefly: 3*2=6

(i) GPU

(ii) VLIW

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